FAST NUCES, Islamabad Campus

Granite

Find the source of silent AI failures.

Registered: A Self-Healing Multi-Agent System
for Semantic Error Diagnosis, Verified Repair and Recovery

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Project goals

- Detect wrong answers
- Locate their source
- Verify proposed fixes

For engineers & QA teams

F26-060 · BS (DS) · FYP 2026

AI agents can fail silently.

× **Agent reply**

$662.00

Check the facts →

✓ **Catalogue total**

$662.23

Refund Desk · recorded run, no error raised.

Find the cause. Test the fix.

Implemented

AI workflow
Record agent steps

→

Check
Test against trusted data

→

Trace
Find the error's origin

→

Replay
Compare fix and control

→

Human review
Accept or reject

Evidence stays with every decision.

Test with and without the fix.

Corrected source value

With fix	✓ ✓ ✓ ✓ ✓	5/5 correct
No fix	× ×	0/2 correct

10/10
faulty runs detected

9/10
origins verified

Feasibility pilot · 7 Oct 2026
10 tasks; one fault; one small model; four tasks not blind.

FYP-1 plan · tentative iterations

Built
1 · Investigate
Refund Desk workbench
Checks · trace · replay

→

Planned
2 · Guide repairs
One-click investigation
Derived fixes · Fault Lab

→

Planned
3 · Extend
Retail Agent + imports
Baselines + test export

Tools & technologies

Python · FastAPI · React · TypeScript · Ollama